doi:10.1093/bjc/azi054 BRIT. J. CRIMINOL. (2006) **46**, 258–285
Advance Access publication 14 June 2005

CRIME MEASURES AND THE SPATIAL ANALYSIS OF CRIMINAL ACTIVITY

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This paper investigates the spatial aspect of criminal activity in Vancouver, Canada, employing social disorganization theory, routine activity theory and multiple measures of crime. Crime counts and crime rates with residential and ambient populations as denominators are calculated using the calls for service made to the Vancouver Police Department. The ambient population—a 24-hour average estimate of a population in a spatial unit to capture the population at risk—is obtained from the LandScan Global Population Database and calculated at a spatial resolution relevant to criminological research. Utilizing a spatial regression technique, strong support is found for routine activity theory across space and the use of ambient populations when calculating crime rates and measuring the population at risk.

I. Introduction

The measurement of crime is a necessity for any quantitative assessment of criminological theory or policy change. This measurement is particularly useful when comparing two or more spatial units, with respect to criminal activity. Crime counts are one measure of criminal activity often used in the spatial analysis of crime. These counts are typically used in 'hotspot' analysis, with hotspots generally being defined as areas with a 'higher than expected level of criminal activity' (Ratcliffe and McCullagh 2001: 331). However, high crime counts may simply be present because of the volume of people in a given area. Because crime occurs when an offender, a victim and a law intersect in time and space, a greater number of people implies a greater number of both potential offenders and potential victims.

Given that high levels of crime may occur simply because of the high levels of people in particular places such as central business districts (Schmid 1960*a*; 1960*b*), crime rates are commonly used to assess the risk of crime by controlling for the population at risk—this property constitutes the crime rate as one of the most common measures in crime analysis, usually employing the residential population as the control. However, crime rates are not without their limitations. As shown by Boggs (1965), changing the population at risk (using residential population versus the number of automobiles present in an automotive theft crime rate, for example) may substantially change the pattern of crime. Extending Bogg's (1965) analysis of changing denominators in crime rates, Harries (1991) finds that although different denominators may add additional information to an analysis, an inappropriate application may be misleading and potentially expensive in both research funds and time allocation. Therefore, the costs and benefits of alternative denominators must not be taken lightly.

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This paper employs a relatively new measure of the population at risk—the ambient population. Calculated by the Oak Ridge National Laboratory, the ambient population measures the number of people that can be expected in an area at any given time of the day or year—these data are free for non-commercial use. The novel aspect of these data, aside from cost, is their spatial scale—approximately one square kilometre. Previous ambient or daytime population estimates are at a scale too coarse for the spatial analysis of crime in an urban centre.

In order to assess which crime measure is most appropriate for an analysis, the statistical results from the use of crime counts, crime rates using residential populations, crime rates using dwelling counts (break and enter) and crime rates using ambient populations are compared using social disorganization and routine activity theory. If the results from the ambient-based calculations are not qualitatively different from the crime counts and/or crime rate calculations using the more common residential-based populations, little utility is gained from the extra effort to obtain ambient population data. However, if the results are qualitatively different, past evaluations of criminological theory and policy may be called into question.

The remainder of the paper is organized as follows: section II reviews the literature on social disorganization theory, routine activity theory and their theoretical synthesis; section III discusses the data and methodology employed; section IV presents the results and discussion; and section V concludes that the ambient population estimates are important to incorporate into a spatial analysis of criminal activity.

II. Social Disorganization and Routine Activity Theory

The spatial, or environmental/ecological, approaches to the theory of crime date back to the early nineteenth century—see Guerry (1833) and Quetelet (1842). More modern work using a spatial perspective on crime has its roots in the work of Shaw (1929) and Shaw and Mackay (1942), as well as research that began in the late 1960s and early 1970s, culminating in the publication of Jeffery (1971) and Newman (1971). The point of departure for the research in this genre is that human behaviour is situated in place; therefore, the place in which crime occurs needs to be (at the very least) one of the dimensions of crime investigated—essentially, crime has a geography. Within the geography of crime and criminological research, there are two dominant spatial theories of crime relating back to this early work: social disorganization theory and routine activity theory.

Not surprisingly, given the spatial nature of these two theories of crime, efforts have been made to integrate them into empirical analyses of crime. Gottfredson *et al.* (1991), Sampson and Wooldredge (1987), Simcha-Fagan and Schwartz (1986) and Smith *et al.* (2000) have all shown that variables representing both social disorganization and routine activity theories prove to be good predictors of various victimization rates. Therefore, this hybrid approach is undertaken here.

In order to capture social disorganization, variables representing ethnic heterogeneity, social/economic deprivation, family disruption and population turnover are employed (see Cahill and Mulligan 2003; Linsky and Straus 1986; Sampson and Groves 1989; Stark 1996; Tseloni *et al.* 2002): ethnic heterogeneity is measured using the percentage of recent (1981–96) immigrant population per spatial unit; social/economic deprivation is captured by the unemployment rate, the percentage of people with a

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university degree, average family income and the standard deviation of average family income in the spatial unit; family disruption is measured with the percentage of single-parent families; and population turnover is measured through population change and the percentage of rental residences in the spatial unit. Ethnic heterogeneity, unemployment, single-parent families, population change, the standard deviation of average income and rental residences are all expected to have positive relationships with crime rates, whereas education and income are expected to have negative relationships with crime rates.

Unlike social disorganization theory, which is defined at the neighbourhood level, variables representing routine activity theory are usually gained through victimization surveys. This research breaks from that practice. Though victimization surveys contain detailed accounts of criminal victimization, many of the foundational ideas of Cohen and Felson (1979) can be measured at the census tract level: income, employment, activities, etc. are all easily represented through variables in the census—these measures, of course, can only be interpreted as averages for the census tract. In fact, I would argue, they are a better representation of routine activity theory than victimization surveys.

Victimization surveys undertaken as part of a research project, due to their cost, only report the activities of people that have actually been victimized. With no control group (non-victims of crime), there is no way of knowing whether there is anything special or peculiar with the activities of the victims versus the non-victims of crime. National victimization surveys, in contrast, contain information on a sample of the total population (approximately 70,000 in the United States and 25,000 in Canada). Information from these surveys is gathered on both victims and non-victims of crime for the purposes of inference. This sampling technique alleviates the difficulties associated with not having a control group, but introduces another, geographical, difficulty—the ecological fallacy. Any activities inferred at the national level must necessarily be assumed to be representative of the local area under study if these national victimization surveys are to be used. However, as averages for the nation, such averages are typically in error for local studies—what is true of the whole is not necessarily true of its parts.

Despite instructive analyses using victimization surveys (see Tseloni *et al.* 2004), in its essence, routine activity theory is about the increased *potential* of criminal victimization as a result of *any* person's activities. To capture this potential, the researcher can measure routine activity variables for neighbourhoods across space because the same variables that Cohen and Felson (1979) cite as not being constant over time are also not constant over space within the same time period. Different neighbourhoods have different age, ethnicity and marital status characteristics (the most powerful predictors of victimization found in empirical applications of routine activity theory) and, therefore, different neighbourhoods have different routine activities, on average.

The variables that affect routine activities, and thus targets and guardianship, revolve around populations; employment status, including the number of hours away from home; income levels; and dwelling values (Cohen and Felson 1979; Clarke and Felson 1993; Fisher and Wilkes 2003; Tseloni *et al.* 2004). In order to measure guardianship in a spatial unit of analysis, or lack thereof, population density is used in the statistical model below. Increases in the population size are expected to increase the number of potential targets/victims of crime, having a positive effect on crime, but an increase in

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population density, holding all other variables in the model constant, is expected to have a negative relationship with crime through increased guardianship/protection—potentially explaining the negative relationship with violent crime found in Cahill and Mulligan (2003). Suitable targets/victims are represented with the population size, number of dwellings, average family income, average dwelling value and percentage of rental residences. The number of dwellings increases the number of targets, whereas family income and dwelling value both increase the suitability of those targets. Increases in family income also represent increases in disposable income and the percentage of rental residences are expected to lead to more activities away from the home. These variables are expected to have a positive effect on crime measures. Finally, unemployment and percentage of the population being prime offending age (15–29) are used to capture economic deprivation and the presence of offenders, respectively. This should be particularly true for property crime. Therefore, a positive effect is expected between unemployment and crime rates, as well as the percentage of young populations and crime rates.

Clearly, there is substantial overlap in the variables representing these two theories, some with conflicting prior expectations; increases in income decreases social disorganization and, therefore, crime, but those same increases also increase the suitability of being a target of victimization, increasing crime. As such, the integration of these two spatial theories of crime adds to the literature in isolating the independent effects of variables on crime rates to identify which theory each variable is best associated with.

III. Data and Methodology

Socio-demographic and socio-economic data

The socio-demographic and socio-economic variables used to represent social disorganization and routine activity theories come from Statistics Canada's 1996 Census data, distributed by DMTI Spatial Inc. A focused neighbourhood analysis of crime using census data of a different year may obscure relationships between crimes and socio-demographic and socio-economic variables (Ley 2004). Therefore, given that the crime data cover the year 1996, the same year of census data is employed.

The descriptive statistics and correlations for these variables for Vancouver, mentioned in the previous section, are provided in Tables 1 and 2, respectively. None of the correlations in Table 2 shows any surprising relationships, with the highest of these correlations involving the income and education variables. Average family income has a strong positive relationship with the average dwelling value, university education and the standard deviation of average family income—these are all standard relationships. Strong negative relationships are present between university education, the unemployment rate and ethnic heterogeneity. Again, these relationships are standard, given that a university education increases employability and recent immigrants may be less likely to have a university education—notable exceptions to this relationship between ethnic heterogeneity and university education exist, but are present in particular places within Vancouver and tend to be associated with the most recent of immigrant populations. With regard to the spatial statistical modelling below, some of the correlations cause concern for collinearity. None of the correlation coefficients is greater than 0.80—a common threshold for concern—but a number of the correlation coefficients are high.

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TABLE 1 *Descriptive statistics for independent variables*

	Mean	Standard deviation	Minimum	Maximum
Ethnic heterogeneity	24.3	9.1	4.7	44.0
Unemployment rate	9.94	4.65	3.2	33.1
University degree, %	20.8	11.5	3.8	46.2
Population change, %	10.79	19.64	−3.90	141.1
Single parents, %	16.2	5.3	6.3	31.7
S.D. average income, 000s	3.4	3.6	1.1	23.8
Average income, 000s	58.7	25.1	18.9	170.1
Rental residences, %	51.7	20.3	16.8	97.7
Population, 000s	5.9	1.8	1.5	12.0
Population density, 000s	6.3	4.8	1.0	26.8
Number of dwellings, 00s	25.48	11.16	7.75	61.38
Average dwelling value, 000s	386.81	190.11	126.49	1088.75
Young population, %	23.95	4.31	10.26	35.84
Ambient population, 000s	5.03	0.303	0.31	15.68
Ambient population density, 000s	4.16	0.182	1.28	9.53

Source: DMTI Spatial Inc. and Oak Ridge National Laboratory; calculations by the author.

In order to avoid the ecological fallacy in the socio-demographic and socio-economic variables, as well as their corresponding crime measurements, smaller spatial units are superior to larger spatial units, due to less averaging of the data. In Statistics Canada's Census of 1996, the smallest spatial unit is the enumeration area, followed by the census tract—in the City of Vancouver, there are 860 enumeration areas and 87 census tracts. However, in order to obtain certain socio-demographic and socio-economic variables for the statistical modelling below, there must be a minimum of 250 residents in the census unit for reasons of confidentiality. Of the 860 enumeration areas in Vancouver, 211 (or 24.5 per cent) have fewer than 250 residents and, therefore, do not have the corresponding socio-demographic and socio-economic variables. If these enumeration areas are located randomly across space, there is no difficulty with performing the analysis, but, as shown in Figure 1, these enumeration areas are clustered in the Downtown Eastside of Vancouver. This is a region of high crime, by most measures, in Vancouver and would necessarily have to be excluded in any analysis due to the lack of socio-demographic and socio-economic variables. To ease this difficulty, crime is measured at the census tract scale, all of which have the necessary socio-demographic and socio-economic information.

Measuring the population at risk

As stated above, alternative measures of the population at risk have been advocated since the 1960s and shown to have significant implications for the patterns of criminal activity (Boggs 1965; Harries 1991). However, the costs of obtaining better measures of the population at risk come at both high monetary and time costs. Now, due to the availability of age-, gender- and ethnic-based demographic data from the census, the reporting of offender-based crime rates through national statistical agencies and academic research has begun to emerge (see Fox and Piquero 2003; Hayslett-McCall and Bernard 2002; O'Brien and Stockard 2003; Wooldredge and Thistlewaite 2003). Until recently, and despite the known limitations of using residential populations collected

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TABLE 2 *Correlations for independent variables*

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}	X_{11}	X_{12}	X_{13}	X_{14}	X_{15}
X_1	1														
X_2	0.306	1													
X_3	−0.701	−0.688	1												
X_4	−0.075	0.100	0.058	1											
X_5	0.092	0.589	−0.473	−0.125	1										
X_6	−0.194	−0.318	0.491	0.053	−0.395	1									
X_7	−0.281	−0.594	0.710	0.021	−0.595	0.788	1								
X_8	−0.469	0.292	0.157	0.176	0.315	−0.078	−0.320	1							
X_9	0.181	0.113	−0.219	0.048	0.157	−0.417	−0.281	−0.013	1						
X_{10}	−0.302	−0.069	0.111	−0.037	−0.003	−0.011	−0.211	0.541	−0.009	1					
X_{11}	−0.387	0.102	0.197	0.255	0.094	−0.199	−0.194	0.638	0.627	0.408	1				
X_{12}	0.157	−0.435	0.356	−0.140	−0.555	0.514	0.772	−0.629	−0.205	−0.485	−0.474	1			
X_{13}	−0.226	−0.376	0.344	0.260	−0.309	−0.093	0.034	0.271	0.131	0.370	0.330	−0.141	1		
X_{14}	0.220	0.204	−0.212	−0.092	0.104	−0.021	0.030	−0.199	0.411	−0.535	0.078	0.186	−0.197	1	
X_{15}	−0.036	0.238	−0.159	0.038	0.278	−0.059	−0.187	0.295	−0.042	0.102	0.068	−0.264	0.220	0.339	1

X_1, ethnic heterogeneity; X_2, unemployment rate; X_3, university degree, %; X_4, population change, %; X_5, single parents, %; X_6, S.D. average income, 000s; X_7, average income, 000s; X_8, rentals, %; X_9, population, 000s; X_{10}, population density, 000s; X_{11}, number of dwellings, 00s; X_{12}, average dwelling value, 000s; X_{13}, young population, %; X_{14}, ambient population, 000s; X_{15}, ambient population density, 000s. Source: DMTI Spatial Inc. and Oak Ridge National Laboratory; calculations by the author.

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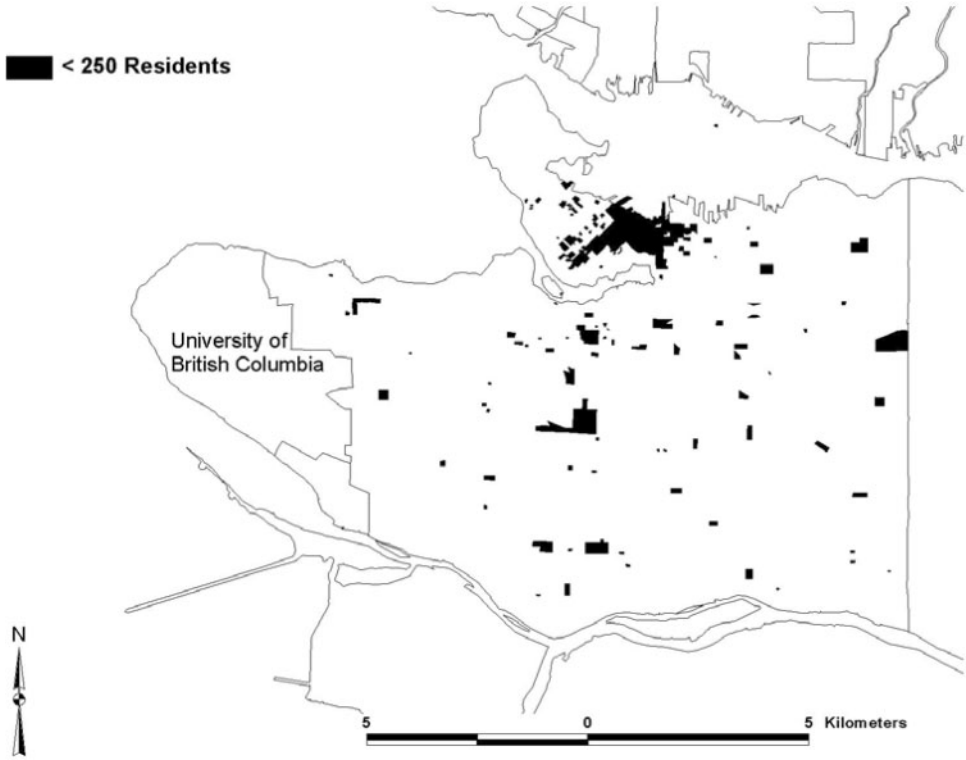

FIG. 1 Enumeration areas with low population.
Source: DMTI Spatial, Inc.

by the census (Miller 2000), the most reliable and cost-effective measures of the population at risk has been the census residential population of the spatial unit under study. But in order to best measure the population at risk, the science of knowing where people actually are is critical (Dobson 2004).

The LandScan Global Population Database has been adopted by many US and international government agencies, as well as the United Nations, and become the 'de facto world standard for these government bodies for estimating populations at risk from terrorism, technological accidents, regional conflicts, and natural disasters' (Dobson 2003: 162). The uses of these data in areas outside of the potentially large-scale disasters are vast, including research within criminology. The resolution of the LandScan database is 30 arc seconds by 30 arc seconds (latitude and longitude), which is approximately one square kilometre—a 'quantum leap' in the precision of population estimates from the previous world standard, P-95 and Rural Cell Population from the US Census Bureau (Dobson 2003: 163).

The LandScan Global Population Database provides a 24-hour ambient population estimate. It is an estimate of how many people are in a given square kilometre at any time of the day and all days of the year because the measure incorporates diurnal and

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seasonal changes. Throughout the day, people move away from their residences to work, shop and recreational activities. The LandScan database assigns each one square kilometre cell a value based on its relative attractiveness to people pursuing these activities. The population within a nation, based on census counts, is then redistributed based upon each cell's attractiveness to the population (Dobson *et al.* 2003). The resulting database is a population estimate at the one kilometre by one kilometre scale that 'integrates diurnal movements and collective travel habits into a single measure' and its 'purpose is to distribute populations based on their likely ambient locations integrated over a 24-hour period for typical days, weeks, and seasons' (Dobson *et al.* 2000: 849–50).

The census counts are redistributed to the cells using a probability coefficient based on road proximity, slope, land cover and night-time lights. Transportation networks (roads, rail, water and air) are all excellent indicators of populations, particularly road network densities, which is positively related to population density; slope enters the calculations because human settlements are typically located on relatively flat terrain; land cover (desert, water, wetlands, ice, urban, etc.), due to suitability of human settlement, is an important factor (Dobson *et al.* 2000); and '[n]ighttime lights are the best available global indicator of where people live, work, and play, and the amount of light emitted is roughly proportional to the number of people' (Dobson 2004: 577).

The overall model is the same for all regions of the world, but the individual variables need to (and do) represent different places differently—the presence of night-time lights in energy-rich nations has different implications than for energy-poor nations. Census counts are used for population control totals, usually at the provincial level (one administration division below national). Additionally, verification and validation studies are conducted routinely for all regions of the world, and intensely for some regions within the United States (Dobson *et al.* 2000).

The differences between the ambient and census residential populations are, at times, extreme. In a downtown region of Houston, Texas, with a residential population of approximately 7,000, its ambient population estimate is approximately 180,000—off by a factor of 26 (Gold 2003). Needless to say, the uses of these data for crime rate calculations are manifold. Such radical differences are not present in Vancouver, but the utility of employing the ambient population estimates is apparent below.

Crime data

The data measuring criminal activity within Vancouver is the Vancouver Police Department (VPD) Calls for Service Database. The calls for service are telephone calls requesting police service that have been made directly to the VPD or through the 911 Emergency Service and allocated to the VPD. Information provided includes the geographic location, date, time of day, complaint code/description, unit response time, priority code, etc. Due to the vast quantity of data available—approximately 30,000 calls for service each month (see Table 3)—only certain crimes are investigated. The advantage of these data is their raw form. These are not the official crime statistics from Statistics Canada that are dependent on a criminal charge, but record actual police activity across the city. Consequently, these data provide a richer assessment of criminal activity in Vancouver. One further data selection criterion is necessary to map criminal activity for Vancouver: which criminal activities to map.

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TABLE 3 *Monthly frequencies of Vancouver Police calls for service, 1996s*

	Frequency	Percentage	Cumulative percentage
January	27,501	7.91	7.91
February	27,039	7.78	15.69
March	28,488	8.2	23.89
April	28,702	8.26	32.15
May	29,888	8.6	40.75
June	30,301	8.72	49.47
July	31,591	9.09	58.56
August	32,738	9.42	67.98
September	28,812	8.29	76.27
October	29,388	8.46	84.72
November	27,824	8.01	92.73
December	25,263	7.27	100
Total	347,535	100	

Source: Vancouver Police Department; calculations by the author.

Criminology groups crime into three broad classifications: violent crime, property crime and nuisance/other crime (Ellis and Walsh 2000). Nuisance crimes such as noise complaints and suspicious (but unconfirmed) activity are not investigated because crimes against person and property have a significantly greater qualitative impact. Violent crime includes assault, fighting, hold-ups, homicide, robbery, sexual assault and stabbing. Property crime is broken into two classifications: break and enter, that includes actual and attempted break and enter; and automotive theft, which includes stolen automobiles and items stolen from automobiles—these categories also have a classification for the crime being in progress. The resulting data-set includes 11,991 violent crimes, 23,262 break and enters, and 36,720 automotive thefts.

Geocoding the point locations of these crimes provides potential for error. Aside from the accuracy of geocoding algorithms (see Ratcliffe 2001), geocoding procedures run the risk of not finding the addresses of point locations. Though the imported data file must be in a format recognized by the GIS software and street addresses must be standardized, particular addresses or point locations may not be mapped. The geocoding procedure used in this paper produced 93, 96 and 94 per cent success rates for automotive theft, break and enter, and violent crime, respectively. After consultation with the proprietors of the data, it was determined that, on occasion, an address is improperly recorded in the field and, therefore, cannot be found by the geocoding procedure—either a non-existent street address is provided or only the street block number. With such a high success rate of perfect matches and the nature of the improper address records appearing to be random, the analysis is performed without concern for bias (Ratcliffe 2004). The final data-set includes 11,848 violent crimes, 22,896 break and enters, and 36,419 automotive thefts.

Measures of crime

Employing and mapping different measurements of crime improves crime analysis by allowing a comparison to be made. In this paper, three measures of crime are used:

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crime counts, crime rates based on residential populations, and crime rates based on ambient populations—crime rates based on the number of dwelling units are also calculated, but only for the crime of break and enter.

Crime counts are a measure of the volume of criminal activity. Though crime counts may be of interest for those investigating where crime 'takes place', crime counts are a restrictive measure of criminal activity because they are in absolute terms—crime counts are absolute, so they do not give any indication of why crime is high in some areas while low in other areas.

Crime rates are commonly used to assess the risk of crime by controlling for the population at risk. Though crime rates are fraught with difficulties of their own (see Andresen *et al.* 2003; Boggs 1965; Harries 1991; Johnson and Lazarus 1989), they are meant to assess the risk of particular people committing or being victimized by particular crimes in particular locations and/or at particular times.

$$\text{Crime Rate per 1000} = \left(\frac{\text{Measure of criminal occurrance}}{\text{Most direct measure of units at risk}}\right) * 1000 \qquad (1)$$

In Equation (1), the numerator is most often a count of a criminal activity, and not contentious. The denominator, on the other hand, is more subjective, as it involves a judgement of the party at risk of offending or being victimized. For example, most crime rates are calculated using the total residential population in the spatial unit under investigation, but only a portion of that population may be at risk of victimization. Alternative denominators can be used in an analysis to provide more meaningful assessments of risk (Boggs 1965), but, historically, the costs of obtaining such data have been high, with the possibility of no significant change in the results (Harries 1991)—the number of dwellings is one alternative for the crime of break and enter that is available through most census data.

In order to incorporate a different and more meaningful assessment of risk and alleviate the difficulties of costs, ambient population estimates are used as alternative denominators in the calculation of crime rates for the analysis below. These data are available free of charge for non-commercial applications and capture the number of people in an area during any time of the day. As such, the risk of victimization is measured with more precision because some places attract many people during the day. A result of this attraction is that these areas may have over-estimates of criminal activity, leading to improper assessments of risk.

Estimation

Originally using an ordinary least squares regression model, the null hypothesis of spatial autocorrelation in the error terms is not rejected. Due to the presence of spatial autocorrelation (this is also evident in the spatial patterns of the figures above), the standard assumption in least squares regression of independent errors is violated (Murray *et al.* 2001). In the presence of positive spatial autocorrelation, present in most socioeconomic data, spatial units contiguous (or even close) to one another have similar values for both dependent and independent variables in a statistical analysis. Therefore, if a positive regression error is calculated for one spatial unit, its neighbours are very likely to also have a positive regression error. This phenomenon allows the researcher

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to predict regression errors spatially, violating the assumption of independence. Though the parameter estimates of the regression model are still unbiased, the variance, and therefore standard deviation, of the errors is not known and standard hypothesis tests are not appropriate—a spatial error model controls for this difficulty.

The general functional form of the spatial error model is as follows:

$$Y = X\beta + \rho \sum_{j=1}^{n} w_{ij}\varepsilon_j + u, \tag{2}$$

where Y is the crime measure, $X\beta$ is a matrix of independent variables and corresponding parameters, w_{ij} is a measure of spatial association between location i and j, ρ is a measure of the strength of that spatial association and u is the remaining independent and identically distributed error. A general-to-specific method is employed for the final model selection: for each crime rate, the statistical model begins with all independent variables, named above; parameters are tested for significance (p-value = 0.05), removed if insignificant; the final models for each crime are reported, displaying the remaining statistically significant independent variables in each case. Spatial autocorrelation is modelled using Queen's contiguity for census tracts. All estimation is performed using GeoDa 0.9.5i—a spatial statistical freeware package developed by Luc Anselin and his co-workers at The Spatial Analysis Laboratory (SAL) in the Department of Agricultural and Consumer Economics at the University of Illinois, Urbana-Champaign (http://sal.agecon.uiuc.edu/geoda_main.php).

IV. Results and Discussion

The descriptive statistics and correlations for the crime measures are presented in Tables 4 and 5. The different rates—residential versus ambient population-based—for the three types of criminal activity exhibit some significant differences. Automotive theft has similar ranges for both the residential-based and ambient-based rates, but the ambient-based average is approximately 133 per cent of the residential-based average. The differences in break and enter are more extreme; the ranges of the different rates

TABLE 4 *Descriptive statistics for dependent variables*

	Mean	Standard deviation	Minimum	Maximum
Auto theft count, Y_1	418.61	92.76	56.00	7,937.00
Break and enter count, Y_2	263.17	24.45	43.00	1,902.00
Violent crime count, Y_3	136.18	30.59	6.00	2,461.00
Auto theft residential rate, Y_4	75.92	15.65	8.88	978.19
Break and enter residential rate, Y_5	44.59	3.64	6.70	229.18
Break and enter dwelling rate, Y_6	103.64	5.70	19.27	309.87
Violent crime residential rate, Y_7	23.04	4.51	1.38	296.54
Auto theft ambient rate, Y_8	109.04	17.05	6.58	1,086.52
Break and enter ambient rate, Y_9	65.88	5.76	7.76	278.75
Violent crime ambient rate, Y_{10}	33.67	5.74	1.14	336.89

All rates are per 1,000 persons. Source: DMTI Spatial Inc., Oak Ridge National Laboratory, and Vancouver Police Department; calculations by the author.

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TABLE 5 *Correlations for dependent variables*

Y_1	1									
Y_2	0.877	1								
Y_3	0.965	0.901	1							
Y_4	0.800	0.630	0.727	1						
Y_5	0.759	0.812	0.767	0.857	1					
Y_6	0.558	0.711	0.564	0.650	0.863	1				
Y_7	0.852	0.777	0.904	0.846	0.889	0.655	1			
Y_8	0.745	0.557	0.710	0.657	0.536	0.241	0.667	1		
Y_9	0.459	0.457	0.466	0.351	0.425	0.215	0.433	0.829	1	
Y_{10}	0.712	0.594	0.770	0.565	0.546	0.281	0.737	0.895	0.758	1
	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7	Y_8	Y_9	Y_{10}

Y_1, automotive theft count; Y_2, break and enter count; Y_3, violent crime count; Y_4, automotive theft residential-based rate per 1,000; Y_5, break and enter residential-based rate per 1,000; Y_6, break and enter dwelling-based rate per 1,000; Y_7, violent crime residential-based rate per 1,000; Y_8, automotive theft ambient-based rate per 1,000; Y_9, break and enter ambient-based rate per 1,000; Y_{10}, violent crime ambient-based rate per 1,000. All Pearson correlations are significant at the 1% level, with the exception of Y_6–Y_8 and Y_6–Y_9, which are significant at the 5% level. Non-parametric analysis produces qualitatively similar results. Source: DMTI Spatial Inc. and Oak Ridge National Laboratory; calculations by the author.

are not too dissimilar, but their averages, as with automotive theft, differ substantially. The residential-based rate is less than half of the dwellings-based rate, with the ambient-based rate in between—all mean rates are statistically different from one another. The violent crime rates exhibit similar differences.

Automotive theft

As shown in Table 5 and Figures 2–4, the three measures of automotive theft are highly correlated across space (spatial randomness is rejected for all three measures) and across measures ($r > 0.65$ for all three measures). All three measures exhibit a concentration of automotive theft in the downtown area (the north-central coastal region of the city), dispersing to the commercial shopping areas and residential areas. However, with a moderately high correlation value ($r = 0.657$), the residential- and ambient-based crime rates are far from perfect substitutes in any empirical evaluation of theory or policy, possibly leading to qualitatively different results.

Turning to the spatial regression results for automotive theft, the two traditional crime measures—the crime count and residential-based rate—do not show any inconsistent results (see Table 6). Both models have a similar R^2, with the residential-based rate model slightly more parsimonious than the crime count model. Considering both the magnitude of the estimated coefficients and the ranges of the independent variables, the unemployment rate has the greatest impact on the dependent variables. When one variable is in both models, the estimated coefficient signs are the same, but there is some variation in variable retention between the two models. In particular, residential population, residential population density and the number of dwellings are present in the crime count model, but absent from the residential-based rate model. This result is not particularly intriguing because the residential-based rate is normalized using the residential population, which is correlated with the number of dwellings. The ambient population, however, is insignificant for both crime counts and residential-based rates,

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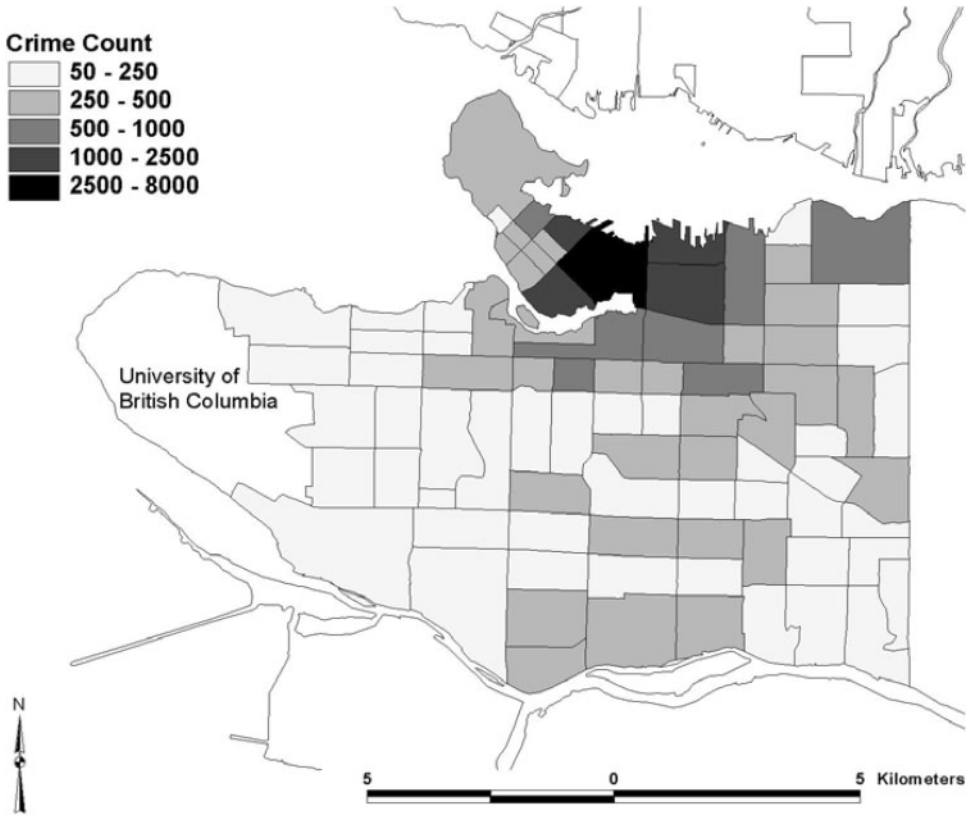

FIG. 2 Automobile theft crime points summed by census tract.
Source: DMTI Spatial Inc. and Vancouver Police Department.

but the ambient-based population density is significant and positive for the residential-based rate. This positive relationship is interpreted as capturing suitable targets for criminal activity; the greater the number of people there are in a given area, the greater number of automobiles there are available to steal or steal from—this finding supports the routine activity approach. Noticeably absent from the crime count and residential-based rate models, though present in the ambient-based rate model, is average family income—this proxy for poverty is typically the strongest predictor of crime in the social disorganization framework.

The ambient-based crime rate results are substantially different from both the crime count and residential-based models. The R^2 is 15–20 per cent higher than the previous two models, with a respectable value of 0.723. The unemployment rate is insignificant, but average family income is present, though positive. These two variables have a moderately high correlation ($r = -0.594$), so multi-collinearity may be the reason why only one of these two variables is present in any model. Overall, the remaining variables are a combination of the variables in the previous two models and the majority of the estimated coefficients are qualitatively similar. Only residential

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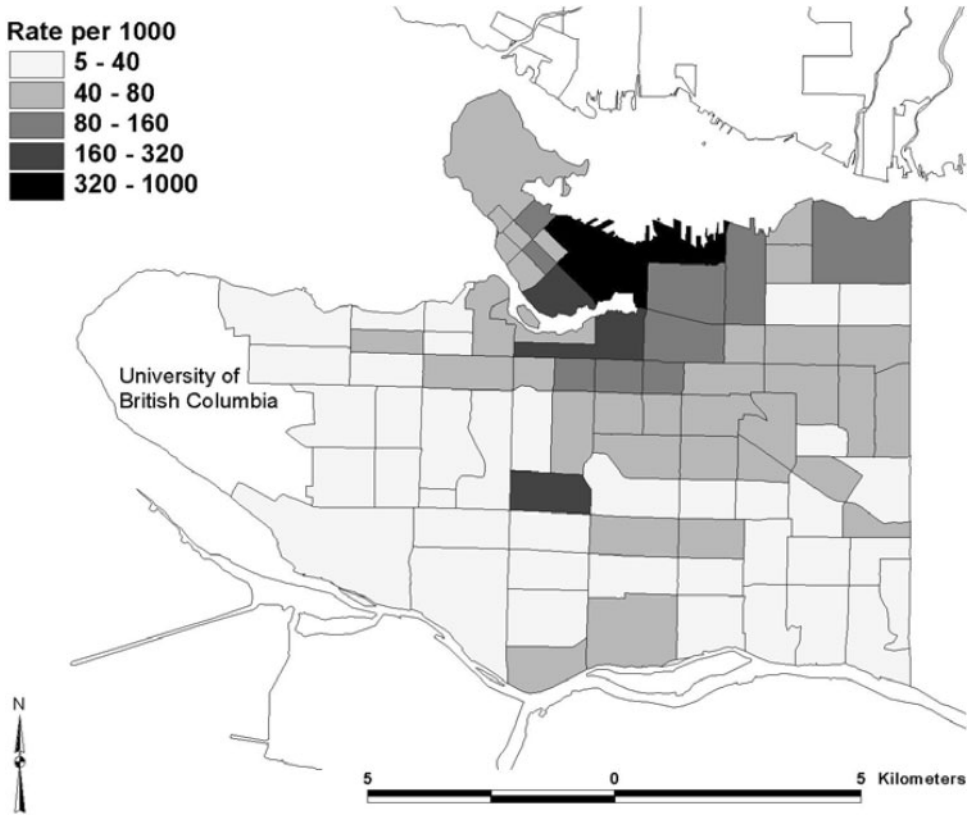

FIG. 3 Automobile theft crime rate per 1,000, residential population based.
Source: DMTI Spatial Inc. and Vancouver Police Department.

population density and ambient population density have the opposite sign of the previous results.

The most striking difference is the opposite sign for ambient population density because this variable is present for both of the calculated rates—the opposite sign for residential population density is in regard to the crime count that may be a result of only one variable being normalized. Curiously, increases in the ambient population density decreases the ambient-based crime rate. Therefore, even when the ambient population is controlled for in the ambient-based crime rate, ambient population density still decreases the automotive theft rate. Independently of the residential-based crime rate results, this finding may be interpreted through routine activity theory as the result of guardianship. Of course, the presence of a high population density does not directly imply guardianship because, as shown in the residential-based rate results, it may also imply an increase in the number of targets and increase crime. The main difficulty with interpreting this result is the absence of the ambient population in both models. The *a priori* expectation is that the population increases targets and crime, but the population density increases guardianship and decreases crime—the low correlation

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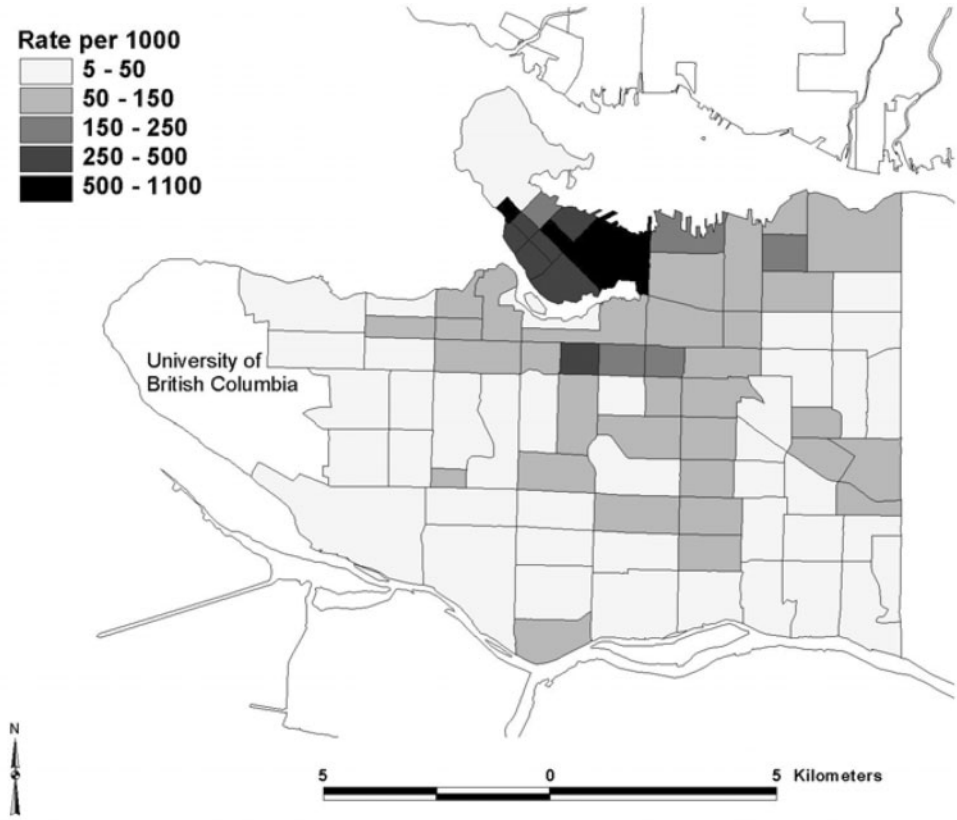

FIG. 4 Automobile theft crime rate per 1,000, ambient population based.
Source: DMTI Spatial Inc., Oak Ridge National Laboratory, and Vancouver Police Department.

TABLE 6 *Spatial regression results: automotive theft*

	Crime count	Residential rate	Ambient rate
Ethnic heterogeneity			
Unemployment rate	99.24(<0.001)	17.51 (<0.001)	
University degree, %		5.78 (<0.001)	18.24 (<0.001)
Population change, %	8.12 (0.017)	3.43 (<0.001)	2.61 (<0.001)
Single parents, %	−42.78 (0.003)		−4.64 (0.038)
S.D. average income, 000s			
Average income, 000s			1.15 (0.019)
Rentals, %			
Population, 000s	−133.47 (0.003)		−25.94 (<0.001)
Population density, 000s	−45.78 (0.003)		16.14 (<0.001)
Number of dwellings, 00s	52.08(<0.001)		5.89 (<0.001)
Average dwelling value, 000s			
Young population, %			
Ambient population, 000s			
Ambient population density, 000s		16.84 (0.009)	−15.81 (0.003)
Spatial dependence, *LR*	0.57 (0.45)	0.02 (0.89)	2.09 (0.15)
Weights Matrix Queen's Order	4	1	1
R^2	0.577	0.538	0.723

Note: p-values are reported in parentheses.

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of these two variables for both residential and ambient measures indicates that population and population density are capturing different phenomena. Although the residential population and residential population density may not give an accurate account of targets or guardianship because they are night-time measures, their estimated coefficients are also the opposite of expectations.

Overall, the ambient-based automotive theft rate does not radically alter the estimated results. The ambient-based rate does retain double the number of variables retained in the residential-based results—the likely reason for almost 20 per cent more variation of the ambient rate being explained by the independent variables. Additionally, the ambient population is a better estimate of the number of people in an area than the residential population, given that most people leave their 'home' census tract during the day. Knowing how many people are in an area is a better indicator of the number of vehicles in that area being vulnerable to theft.

Break and enter

The maps of the four measures of break and enter are shown in Figures 5–8. The crime count, residential-based rate and dwelling-based rate are all highly correlated with each other (all $r > 0.70$), indicating that these three different measure are relatively substitutable—this is particularly true for the residential-based and dwelling-based crime rates ($r = 0.863$). The ambient rate, in contrast, does have positive correlations with the other measures of break and enter, but is clearly measuring a different phenomenon. All correlation coefficients are less than 0.50, with the correlation between the ambient-based rate and the dwelling-based rate being the lowest ($r = 0.215$). Such low correlation values have significant effects on the deployment of police resources. If ambient-based rates are considered relevant for crime prevention efforts by police, these efforts may be concentrated in the wrong neighbourhoods. All measures of break and enter also strongly reject spatial randomness.

The spatial regression results for break and enter (see Table 7) are somewhat different from those for automotive theft. The crime count, residential-based rate and dwelling-based rate all show consistent results with respect to the estimated coefficient signs when a variable is present in more than one final model, but the residential-based rate has the greatest variable retention and the R^2 has increased by more than 10 per cent for the crime count and residential-based rate.

The social disorganization variables in these first three crime measures have mixed results. The unemployment rate and population change variables, when significant, have their expected positive coefficient signs, but the percentage of university graduates and the level of family income also have positive coefficient signs, contrary to expectations—the results are similar to those for automotive theft. Though the purpose of this paper is not to test social disorganization theory, yet again, but to compare the statistical results of ambient population crime rates with more traditional measures of crime, social disorganization theory does not bode well in these analyses.

In fact, higher proportions of university graduate and family income levels having a positive impact on crime is completely consistent with routine activity theory. Education and income are highly correlated ($r = 0.71$) and are expected to increase criminal activities due to the increased possibility of the presence of desirable goods such as jewellery

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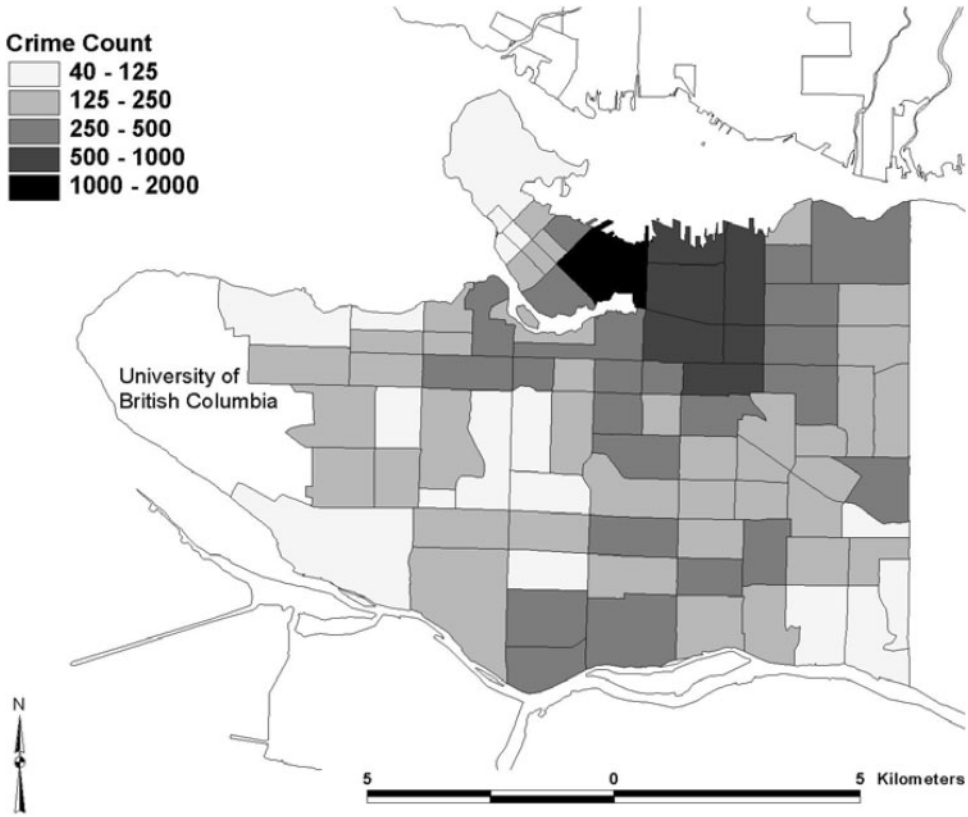

FIG. 5 Break and enter crime points summed by census tract.
Source: DMTI Spatial Inc. and Vancouver Police Department.

and consumer electronics—more potential targets. Also, population change and unemployment, though not typically used in a routine activity framework, are consistent with the expectations of routine activity theory as well. Neighbourhood population turnover increases anonymity, decreasing guardianship, and increased unemployment increases the number of people available to offend or be victims of crime. With the strength of the unemployment rate across all results and the sporadic appearance of average family income, unemployment is a far better predictor of criminal activity than social disorganization theory's claim to poverty as the best predictor, proxied here with income.

Therefore, whether or not people are working appears to be more important than how much income those people have if they work. A person being in a workplace environment necessarily decreases the chances of their being a victim of crime and victimizing someone else. Of course, one's presence at work does not eliminate the potential for criminal activity, but by having other tasks to attend to, the probability of criminal activity decreases. This is a story of potential, which ties back to the routine activity approach.

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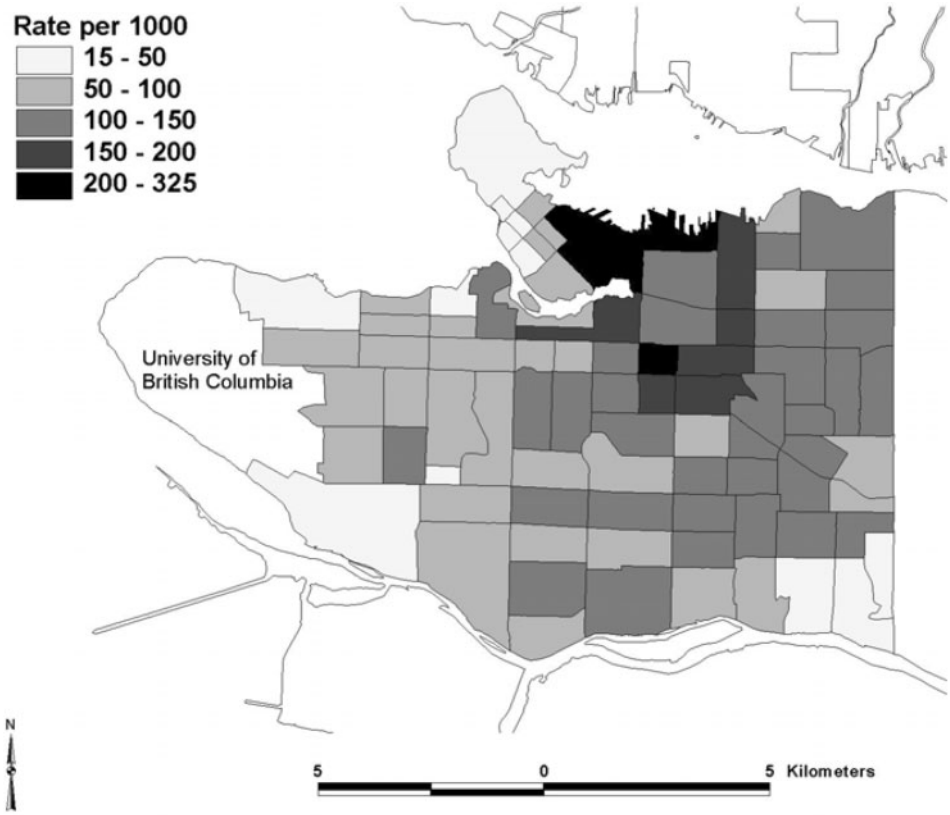

FIG. 6 Break and enter crime rate per 1,000, dwelling count based.
Source: DMTI Spatial Inc. and Vancouver Police Department.

As with automotive thefts, the ambient-based rate for break and enter has the highest R^2 (0.749). Curiously, the lowest R^2 is for the dwelling-based rate results (0.584). The number of dwellings is usually considered the best normalizing variable for break and enter because it objectively measures the number of targets. And if one assumes that the best way to test theory is with the most appropriate measures, the best results should come from that appropriate measure if the theory is correct. Incidentally, the only social disorganization theory variable present in the dwelling-based results is the unemployment rate.

Once again, population and population density exhibit inconsistent results between the ambient-based rate and the other crime measures. The crime count, residential-based rate and the dwelling-based rate all have their expected negative relationships with residential population density; residential population, however, has a negative relationship with the residential-based and ambient-based crime rates; and residential population density has an unexpected positive relationship with the ambient-based rate. The only consistency across all crime measures is the positive impact of the number of dwellings on the measures of break and enter. The ambient population is insignificant for all crime measures. The ambient population density has an unexpected

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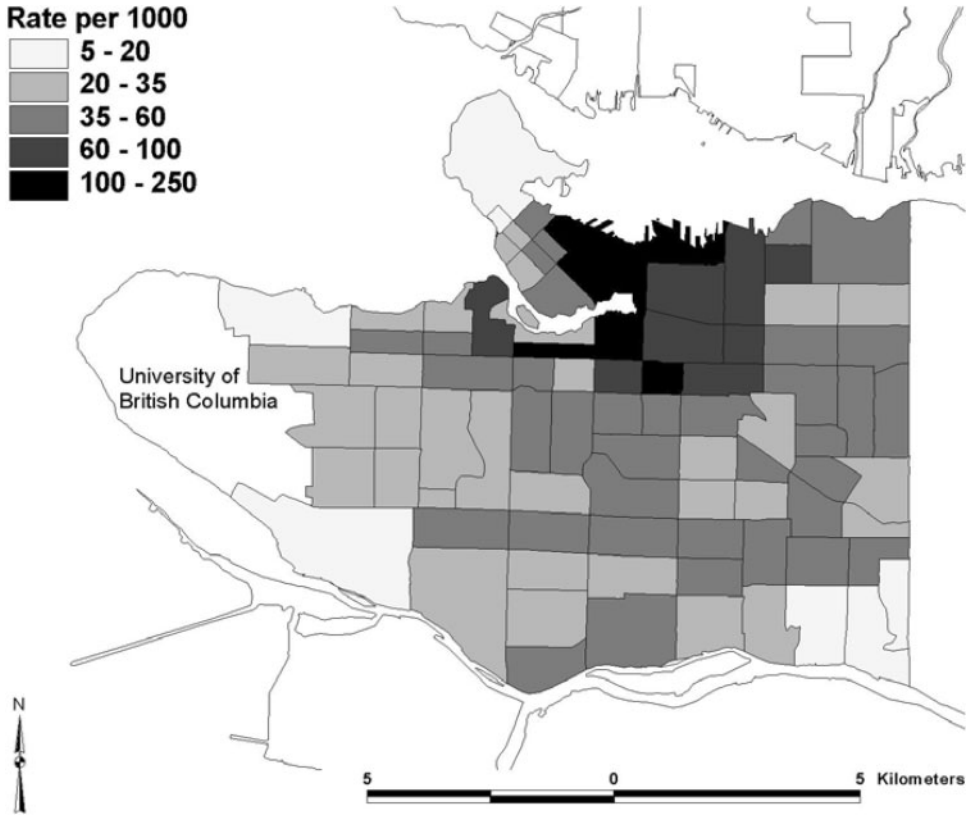

FIG. 7 Break and enter crime rate per 1,000, residential population based.
Source: DMTI Spatial Inc. and Vancouver Police Department.

positive relationship with the residential-based and dwelling-based rates and an expected negative relationship with the ambient-based rate.

The inconsistent results are likely a consequence of improper expectations. The residential and ambient populations are significantly and positively related ($r = 0.411$), but residential and ambient population densities exhibit a relatively low positive relationship ($r = 0.102$). If targets and guardianship are to be captured with population and population density, then the ambient population measures are the most appropriate because they are a better indicator of the number and density of people present at any time of the day, rather than at night. Curiously, the more traditional measures of crime do not correspond to expectations, whereas the ambient-based measure does, with the opposite being true for residential population and residential population density. The question then becomes: what are residential population and residential population density measuring, if not targets and guardianship?

Areas in Vancouver, Canada that typically have high residential populations and, in particular, high residential population densities are apartment and condominium

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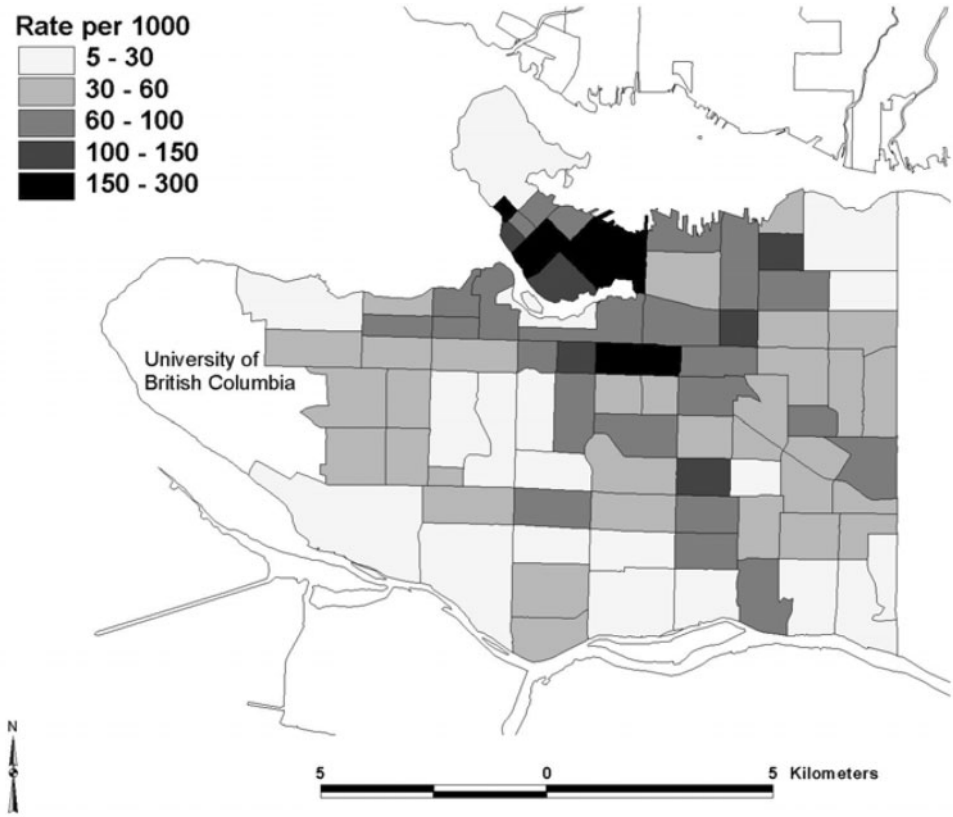

FIG. 8 Break and enter crime rate per 1,000, ambient population based.
Source: DMTI Spatial Inc., Oak Ridge National Laboratory, and Vancouver Police Department.

developments. These developments are typically found in the safer communities of the city and, through secure parking and entrances, provide better protection for automobiles, people and property. This may explain the negative coefficients for residential population and residential population density with the more common measures of crime. With regard to the residential-based crime rate and its positive relationship with the ambient population density, this appears to be a result of the way the crime rate is normalized—similarly for the dwelling-based crime rate.

Using the ratio of the residential to the ambient population as a measure of daytime population attraction—as the ratio increases, fewer people are present in the day than the night—an explanation manifests itself. The areas in Vancouver, Canada that have high residential population densities also have low measures of daytime population attraction (r=–0.566). As a result, any rate calculated with the residential population is artificially low because the normalization variable is in the denominator of the crime rate. Including the ambient population density in the statistical model compensates for that artificially low crime rate with its positive coefficient—this compensation effect is clearly not necessary for the ambient-based rate. Therefore, the differences between the estimated coefficient signs and prior expectations are most likely a result of measurement error in the dependent variable.

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TABLE 7 *Spatial regression results: break and enter*

	Crime count	Residential rate	Dwelling count rate	Ambient rate
Ethnic heterogeneity				
Unemployment rate	31.29 (<0.001)	6.04 (<0.001)	7.35 (<0.001)	5.21 (<0.001)
University degree, %		0.962 (0.004)		
Population change, %		0.275 (0.025)		
Single parents, %				
S.D. average income, 000s				
Average income, 000s	1.41 (0.035)			
Rentals, %				
Population, 000s		−5.07 (0.024)		−6.85 (0.007)
Population density, 000s	−14.23 (<0.001)	−2.07 (0.001)	−4.33 (<0.001)	5.62 (<0.001)
Number of dwellings, 00s	12.42 (<0.001)	0.961 (0.023)		1.37 (0.003)
Average dwelling value, 000s				
Young population, %	8.16 (0.039)	1.49 (0.032)	3.71 (<0.001)	3.77 (<0.001)
Ambient population, 000s				
Ambient population density, 000s		4.61 (0.001)	7.95 (<0.001)	−12.74 (<0.001)
Spatial dependence, *LR*	0.19 (0.66)	1.62 (0.20)	0.29 (0.59)	2.24 (0.13)
Weights Matrix Queen's Order	2	2	1	3
R^2	0.678	0.662	0.584	0.749

Note: p-values are reported in parentheses.

Violent crime

Figures 9–11 show the maps for the three measures of violent crime. As with the two previous crime classifications, violent crime is high correlated across both space and across measures—spatial randomness is strongly rejected for all crime measures and the lowest correlation coefficient is 0.737. As such, violent crime is the most suitable, in the present context, for substitutability between crime measures.

The spatial regression results (see Table 8) exhibit the same consistency between the crime count and the residential-based crime rate. This consistency is even more apparent with the final models all being quite similar. Opposed to the results of automotive theft and beak and enter, the ambient-based crime rate has the lowest R^2 (0.674), with the residential-based rate having the highest (0.777). All of the three final models retain few of the routine activity variables, but the social disorganization theory variables do not correspond well to expectations.

As with the previous results, average family income has a positive relationship with crime, but the violent crime models also include ethnic heterogeneity, the percentage of single-parent households and the percentage of rental residences. All variables have unexpected negative relationships with their respective crime measures. Ethnic heterogeneity in Vancouver, Canada comes dominantly in the form of recent immigrants. In particular, a large proportion of recent immigration has come from Hong Kong and Taiwan, and the majority of these immigrants have high economic status (Ley 1999; Ley and Smith 2000). Therefore, the negative relationship between ethnic heterogeneity and crime rates may not be due only to the underreporting of crime often stemming from English as a second language and a lack of knowledge of (Canadian) law (Ley 2004), but that these immigrants have been settling in the wealthy (i.e. low crime) regions of the city.

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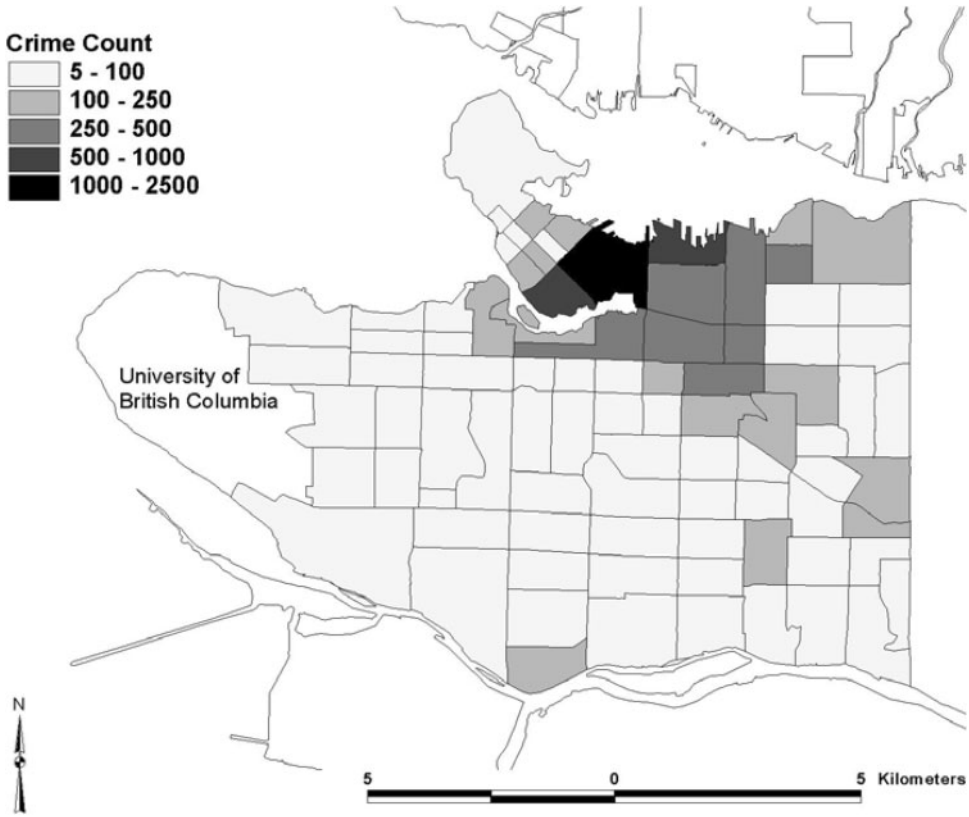

FIG. 9 Violent crime points summed by census tract.
Source: DMTI Spatial Inc. and Vancouver Police Department.

The percentage of single-parent households having a negative relationship with crime rates may not correspond well with social disorganization theory, but does relate to research on intimate partner assault in Canada. DesKeserdy and Ellis (1995) find that once a woman is removed from the residence of her violent partner, but not necessarily the relationship, intimate male violence against that woman decreases. Again, this fits well with routine activity theory—there is a lack of guardianship in the violent male's residence. Of course, not all single-parent households are the result of intimate partner assault, but this example shows that the statistical relationship is only inconsistent with one particular theory, not the existing research.

And finally, we turn to the result for the percentage of rental residences. Given that the majority of rental residences occur in apartment settings in Vancouver, this negative relationship likely corresponds to the explanation above regarding residential population density; the added security of parking and pedestrian entrances adds guardianship to automobiles, property and, with respect to violent crime, people.

Residential population has the same results as the previous crime classifications: negative relationships with the crime count and residential-based crime rate, and a positive

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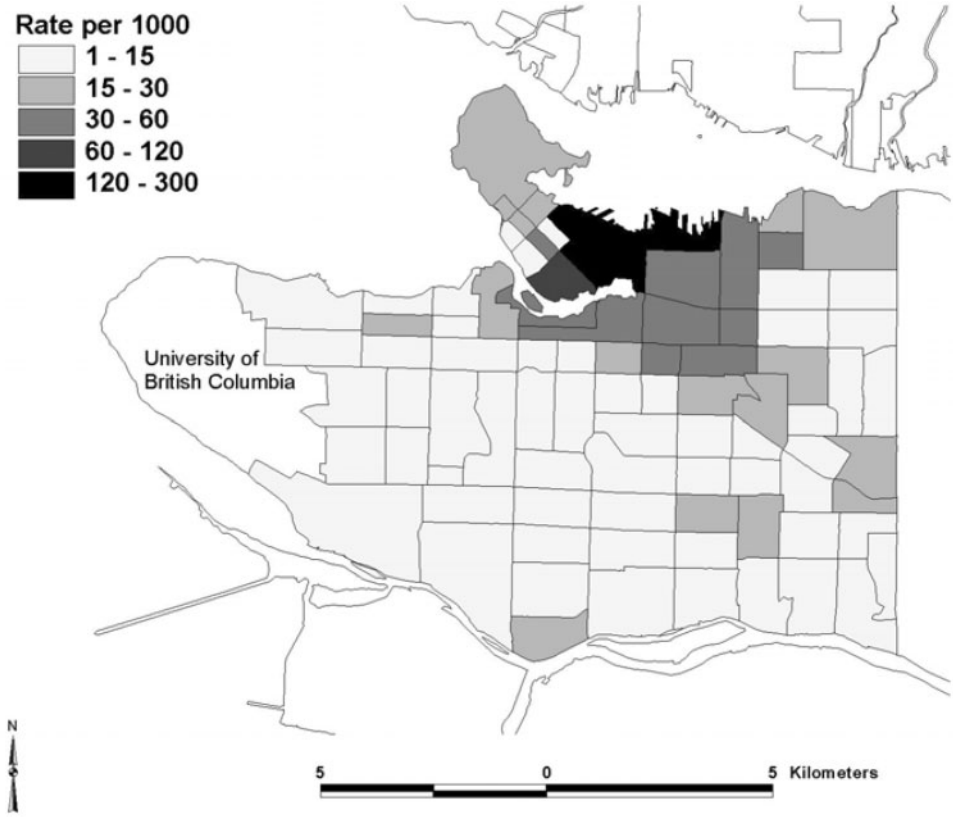

FIG. 10 Violent crime rate per 1,000, residential population based.
Source: DMTI Spatial Inc. and Vancouver Police Department.

relationship with the ambient-based crime rate—the ambient population density is only significant for the ambient-based rate. The number of dwellings positively impacts on violent crime for all these crime measures. Although the number of dwellings is not typically considered in violent crime, this result is likely the manifestation of most violent crime occurring close to the home.

Overall, the results for violent crime are more consistent across crime measures than the previous two crime classifications. The implications for social disorganization theory, however, appear to be worse.

V. Conclusions and Directions for Further Research

This paper uses ambient population estimates from Oak Ridge National Laboratory to calculate crime rates. Because of the nature of these estimated ambient populations, the ambient-based crime rates better assess the population at risk for criminal victimization. Using a spatial statistical technique, as well as social disorganization and routine activity theories as a theoretical framework for automotive theft, break and enter, and

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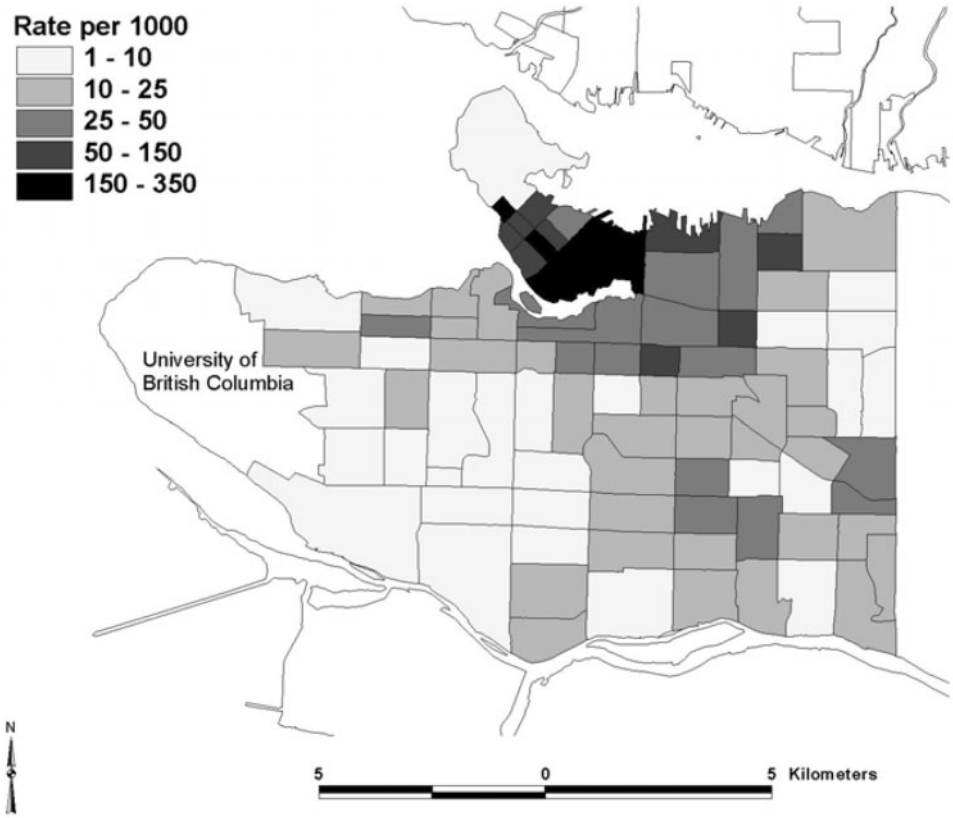

FIG. 11 Violent crime rate per 1,000, ambient population based.
Source: DMTI Spatial Inc., Oak Ridge National Laboratory, and Vancouver Police Department.

violent crime, the results of ambient-based crime rates are compared to the common measures of crime—crime counts and residential-based crime rates.

Crime counts and residential-based crime rates exhibit relatively strong positive correlations for all three crime classifications. The ambient-based crime rates show relatively strong positive correlations among themselves, but less so with the residential-based crime rates—this is particularly true for the crime of break and enter. Therefore, there is no strong case for residential-based and ambient-based crime rates being good substitutes for any of the crime classifications.

Empirically, there are definite differences between the use of the ambient-based crime rate and the other measures of crime. With the exception of violent crime, the ambient population model has superior goodness of fit, potentially due to a better assessment of criminal activity. With respect to the variables representing routine activity theory, there are consistent differences between the ambient-based crime rate results and the other crime measures. These differences are believed to be a result of measurement error in the case of residential-based crime rates and a lack of normalization in the case of crime counts. In addition, this measurement error is apparent for both dependent and independent variables.

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TABLE 8 *Spatial regression results: violent crime*

	Crime count	Residential rate	Ambient rate
Ethnic heterogeneity		−1.17 (<0.001)	−1.18 (0.004)
Unemployment rate	41.36 (<0.001)	7.73 (<0.001)	6.39 (<0.001)
University degree, %			
Population change, %	2.89 (0.001)	0.587 (<0.001)	0.901 (<0.001)
Single parents, %		−1.13 (0.036)	
S.D. average income, 000s			3.37 (0.001)
Average income, 000s	2.06 (0.016)	0.276 (0.019)	
Rentals, %	−4.09 (0.022)		
Population, 000s	−63.12 (0.001)	−5.57 (0.009)	
Population density, 000s	−9.81 (0.029)	−1.32 (0.018)	3.80 (<0.001)
Number of dwellings, 00s	23.39 (<0.001)	1.33 (0.001)	0.891 (0.018)
Average dwelling value, 000s			
Young population, %			
Ambient population, 000s			
Ambient population density, 000s			−6.84 (0.001)
Spatial dependence, *LR*	2.42 (0.12)	0.90 (0.34)	1.82 (0.18)
Weights Matrix Queen's Order	4	2	4
R^2	0.713	0.777	0.674

Note: p-values are reported in parentheses.

Although not an intended result for this paper, social disorganization theory does not fare well in the statistical models. The social disorganization variables that have unexpected signs for their estimated coefficients can be explained using routine activity theory, as can the remaining social disorganization variables that maintain their expectations on the estimated coefficients. Overall, routine activity theory does quite well in explaining criminal activity in Vancouver, Canada.

These results do not provide any major contradictions of past research on the spatial dimension of crime, but does call into question the appropriateness of residential-based crime rates, in general. If unexpected empirical results for the residential-based crime rates can be explained though measurement error, the confirmation or rejection of theories are likely premature. Additionally, the evaluation of public policy using residential-based crime rates may be misinformed. As such, the use of an ambient-based crime rate, if deemed appropriate, should always be employed to avoid measurement error. Past research has indicated that the costs of obtaining better denominators for crime rates may be great given that the changes in the results may not be significant (Harries 1991). However, the financial costs of obtaining these data for non-commercial use is zero and the time commitment is negligible, particularly if research is done jointly with a geographer familiar with geographic information systems.

Future research with ambient populations needs to address a number of issues. First, a similar analysis should be performed at multiple spatial scales of analysis. The empirical application in this paper is limited to the census tract level of analysis because of data availability at a finer resolution, but it is possible that the relationships found are an artefact of the current spatial scale. It should be noted that changing the spatial scale may not only confirm the present results, but provide greater insight. Secondly, the analysis should be performed using finer crime classifications. The question surrounding the number of targets versus guardianship may be resolved using more disaggregated crime data. And finally, previous confirmations and rejections of theories and policy change must be revisited in

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multiple cities and urban centres. The present analysis has found little support for social disorganization theory, despite its empirical success in the past. Incorporating ambient populations in both dependent and independent variables may have severely altered the relationships between socio-demographic/socio-economic variables and crime measures.

Acknowledgements

I am grateful to Greg W. Jenion for his helpful comments on earlier drafts of this paper, as well as two anonymous BJC reviewers for theirs.

References

Andresen, M. A., Jenion, G. W. and Jenion, M. L. (2003), 'Conventional Calculations of Homicide Rates Lead to an Inaccurate Reflection of Canadian Trends', *Canadian Journal of Criminology and Criminal Justice*, 45: 1–17.

Boggs, S. L. (1965), 'Urban Crime Patterns', *American Sociological Review*, 30: 899–908.

Cahill, M. E. and Mulligan, G. F. (2003), 'The Determinants of Crime in Tucson, Arizona', *Urban Geography*, 24: 582–610.

Clarke, R. V. and Felson, M. (1993), 'Introduction: Criminology, Routine Activity, and Rational Choice', in R. V. Clarke and M. Felson, eds, *Routine Activity and Rational Choice*, 1–14. London, UK: Transaction Publishers.

Cohen, L. E. and Felson, M. (1979), 'Social Change and Crime Rate Trends: A Routine Activity Approach', *American Sociological Review*, 44: 588–608.

DeKeseredy, W. S. and Ellis, D. (1995), 'Intimate Male Violence Against Women in Canada', in J. I. Ross, ed., *Violence in Canada: Sociopolitical Perspectives*, 97–125. Don Mills, ONT: Oxford University Press.

Dobson, J. E. (2003), 'Estimating Populations at Risk', in S. L. Cutter, D. B. Richardson and T. J. Wilbanks, eds, *The Geographical Dimensions of Terrorism*, 161–7. New York and London: Routledge.

———(2004), 'The GIS Revolution in Science and Society', in S. D. Brunn, S. L. Cutter and J. W. Harrington, Jr, eds, *Geography and Technology*, 573–87. Dordrecht: Kluwer Academic Publishers.

Dobson, J.E., Bright, E. A., Coleman, P. R., Durfee, R. C. and Worley, B. A. (2000), 'LandScan: A Global Population Database for Estimating Populations at Risk', *Photogrammetric Engineering and Remote Sensing*, 66: 849–57.

Dobson, J. E., Bright, E. A., Coleman, P. R. and Bhaduri, B. L. (2003), 'LandScan: A Global Population Database for Estimating Populations at Risk', in V. Mesev, ed., *Remotely Sensed Cities*, 267–79. London and New York: Taylor and Francis.

Ellis, L. and Walsh, A. (2000), *Criminology: A Global Perspective*. Boston, MA: Allyn and Bacon.

Fisher, B. S. and Wilkes, A. R. P. (2003), 'A Tale of Two Ivory Towers: A Comparative Analysis of Victimization Rates and Risks Between University Students in the United States and England', *British Journal of Criminology*, 43: 526–45.

Fox, J. A. and Piquero, A. R. (2003), 'Deadly Demographics: Population Characteristics and Forecasting Homicide Trends', *Crime and Delinquency*, 49: 339–59.

Gold, S. S. (2003), 'Watch Us Move: Homeland Security Requires a New Kind of Population Map', *Popular Science*, January: 40.

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GOTTFREDSON, D. C., MCNEIL III, R. J. and GOTTFREDSON, G. D. (1991), 'Social Area Influences on Delinquency: A Multilevel Analysis', *Journal of Research in Crime and Delinquency*, 28: 197–226.

GUERRY, A. M. (1833), *Essai Sur la Statistique Morale de la France*. Paris: Crochard.

HARRIES, K. D. (1991), 'Alternative Denominators in Conventional Crime Rates', in P. J. Brantingham and P. L. Brantingham, eds, *Environmental Criminology*, 147–65. Prospect Heights, IL: Waveland Press.

HAYSLETT-MCCALL, K. L. and BERNARD, T. J. (2002), 'Attachment, Masculinity, and Self-Control: A Theory of Male Crime Rates', *Theoretical Criminology*, 6: 5–33.

JEFFERY, C. R. (1971), *Crime Prevention Through Environmental Design*. Beverly Hills, CA: Sage Publications.

JOHNSON, H. and LAZARUS, G. (1989), 'The Impact of Age on Crime Victimization Rates', *Canadian Journal of Criminology*, 31: 309–17.

LEY, D. (1999), 'Myths and Meanings of Immigration and the Metropolis', *The Canadian Geographer/Le Geographe canadien*, 43: 2–19.

———(2004), Personal communication, 20 October 2004.

LEY, D. and SMITH, H. (2000), 'Relations Between Deprivation and Immigrant Groups in Large Canadian Cities', *Urban Studies*, 37: 37–62.

LINSKY, A. and STRAUS, M. (1986), *Social Stress in the United States: Links to Regional Patterns in Crime and Illness*. Dover, MA: Auburn House.

MILLER, J. (2000), 'Profiling Populations Available for Stops and Searches', *Policing and Reducing Crime Unit: Police Research Series Paper 131*. London, UK: Research, Development and Statistics Directorate.

MURRAY, A. T., MCGUFFOG, I., WESTERN, J. S. and MULLINS, P. (2001), 'Exploratory Spatial Data Analysis Techniques for Examining Urban Crime', *British Journal of Criminology*, 41: 309–29.

NEWMAN, O. (1971), *Architectural Design for Crime Prevention*. Washington, DC: Law Enforcement Assistance Administration, National Institute of Law Enforcement and Criminal Justice.

Oak Ridge National Laboratory (2003), *Land Scan Global Population Database*. Oak Ridge, TN: Oak Ridge National Laboratory. Available at http://www.ornl.gov/gist/.

O'BRIEN, R. M. and STOCKARD, J. (2003), 'The Cohort-Size Sample-Size Conundrum: An Empirical Analysis and Assessment Using Homicide Arrest Data from 1960 to 1999', *Journal of Quantitative Criminology*, 19: 1–32.

QUETELET, M. A. (1842), *A Treatise on Man*. Edinburgh: William and Robert Chambers.

RATCLIFFE, J. H. (2001), 'On the Accuracy of TIGER Type Geocoded Address Data in Relation to Cadastral and Census Areal Units', *International Journal of Geographical Information Science*, 15: 473–85.

———(2004), 'Geocoding Crime and a First Estimate of a Minimum Acceptable Hit Rate', *International Journal of Geographical Information Science*, 18: 61–72.

RATCLIFFE, J. H. and MCCULLAGH, M. J. (2001), 'Chasing Ghosts? Police Perception of High Crime Areas', *British Journal of Criminology*, 41: 330–41.

SAMPSON, R. J. and GROVES, W. B. (1989), 'Community Structure and Crime: Testing Social-Disorganization Theory', *American Journal of Sociology*, 94: 774–802.

SAMPSON, R. J. and WOOLDREDGE, J. D. (1987), 'Linking the Micro- and Macro-Level Dimensions of Lifestyle-Routine Activity and Opportunity Models of Predatory Victimization', *Journal of Quantitative Criminology*, 3: 371–93.

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SCHMID, C. F. (1960*a*), 'Urban Crime Areas: Part I', *American Sociological Review*, 25: 527–42.

———(1960*b*), 'Urban Crime Areas: Part II', *American Sociological Review*, 25: 655–78.

SHAW, C. R. (1929), *Delinquency Areas*. Chicago, IL: University of Chicago Press.

SHAW, C. R. and MACKAY, H. D. (1942), *Juvenile Delinquency and Urban Areas: A Study of Rates of Delinquency in Relation to Differential Characteristics of Local Communities in American Cities*. Chicago, IL: University of Chicago Press.

SIMCHA-FAGAN, O. and SCHWARTZ, J. E. (1986), 'Neighborhood and Delinquency: An Assessment of Contextual Effects', *Criminology*, 27: 27–55.

SMITH, W. R., FRAZEE, S. G. and DAVISON, E. L. (2000), 'Furthering the Integration of Routine Activity and Social Disorganization Theories: Small Units of Analysis and the Study of Street Robbery as a Diffusion Process', *Criminology*, 38: 489–523.

STARK, R. (1996), 'Deviant Places: A Theory of the Ecology of Crime', in P. Cordella and L. J. Seigel, eds, *Readings in Contemporary Criminological Theory*, 128–42. Boston, MA: Northeastern University Press.

TSELONI, A., OSBORN, D. R., TRICKETT, A. and PEASE, K. (2002), 'Modelling Property Crime Using the British Crime Survey: What Have We Learnt?', *British Journal of Criminology*, 42: 109–28.

TSELONI, A., WITTEBROOD, K., FARRELL, G. and PEASE, K. (2004), 'Burglary Victimization in England and Wales, the United States and the Netherlands', *British Journal of Criminology*, 44: 66–91.

WOOLDREDGE, J. and THISTLEWAITE, A. B. (2003), 'Neighborhood Structure and Race-Specific Rates of Intimate Assault', *Criminology*, 41: 393–422.

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